

CIRROSE NA UTI

PARTE 4



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Doutor em Ciências Médicas - HCFMUSP



DISFUNÇÃO RENAL X CIRROSE

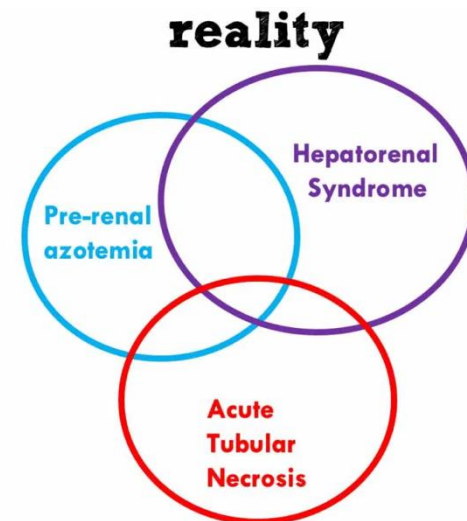


DISFUNÇÃO RENAL X CIRROSE

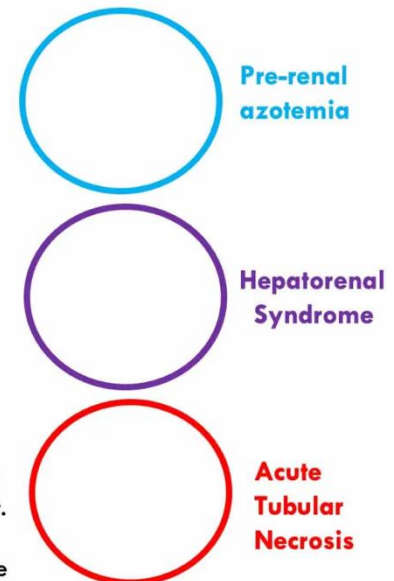
20-50%

dos cirróticos Internados >> IRA

- Sepses
- HDA
- Desidratação
- Drogas nefrotóxicas
- Diuréticos +- Betabloqueadores



literature

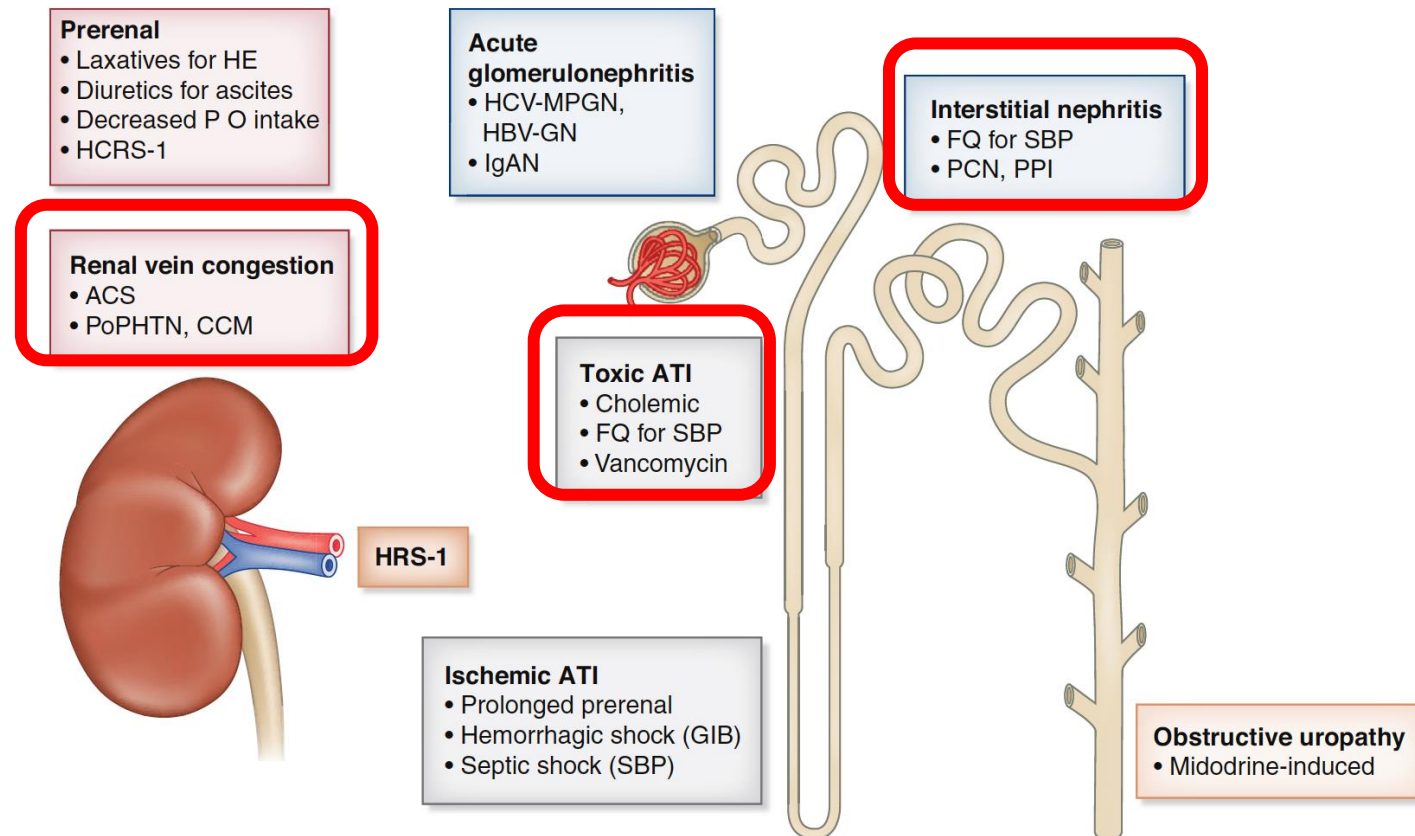


A popular delusion in the medical literature is that patients can only have one cause of kidney injury at a time. This is obviously incorrect. Thus, hepatorenal syndrome is often regarded as a "diagnosis of exclusion," but this narrow view will overlook some patients who have mixed physiological states.

MÚLTIPLAS ETIOLOGIAS ?

Acute Kidney Injury in Patients with Liver Disease

Giuseppe Cullaro ¹, Swetha Rani Kanduri,^{2,3} and Juan Carlos Q. Velez^{2,3}



Terli + Albumina ?



Multicenter Study of Incidence and Outcomes of Acute Kidney Injury including Hepatorenal Syndrome in Hospitalized Patients with Cirrhosis in the United States

HRS-AKI is uncommon and has similar outcomes to acute tubular necrosis.

15 U.S. hospitals from the
HRS HARMONY Consortium

Consecutive patients
admitted with AKI and
cirrhosis in 2019

90-day outcomes with
adjudicated etiology of AKI

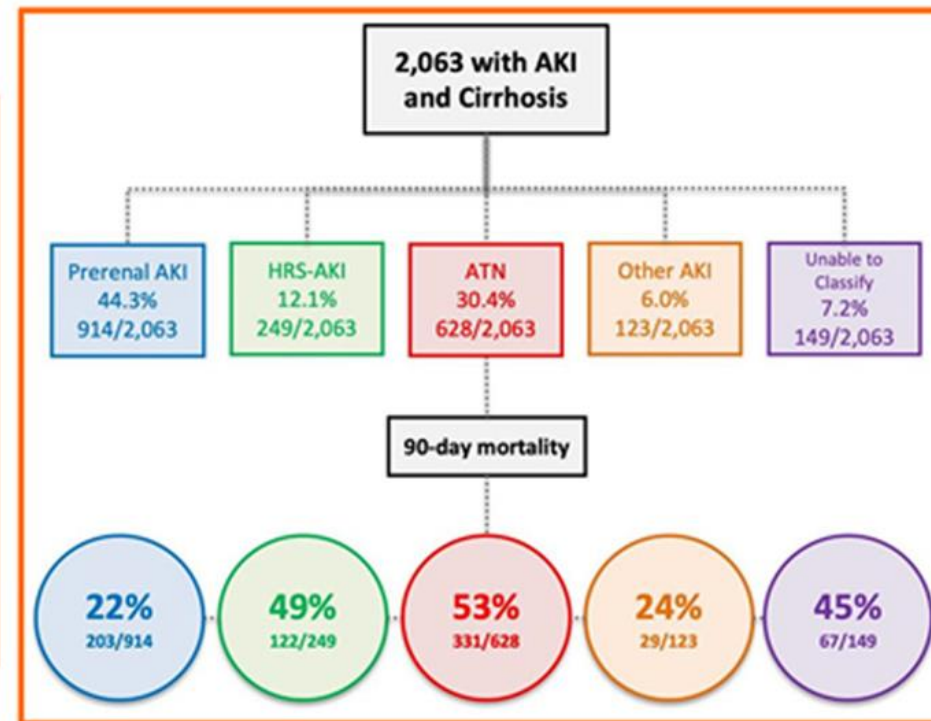


MELD-Na 26

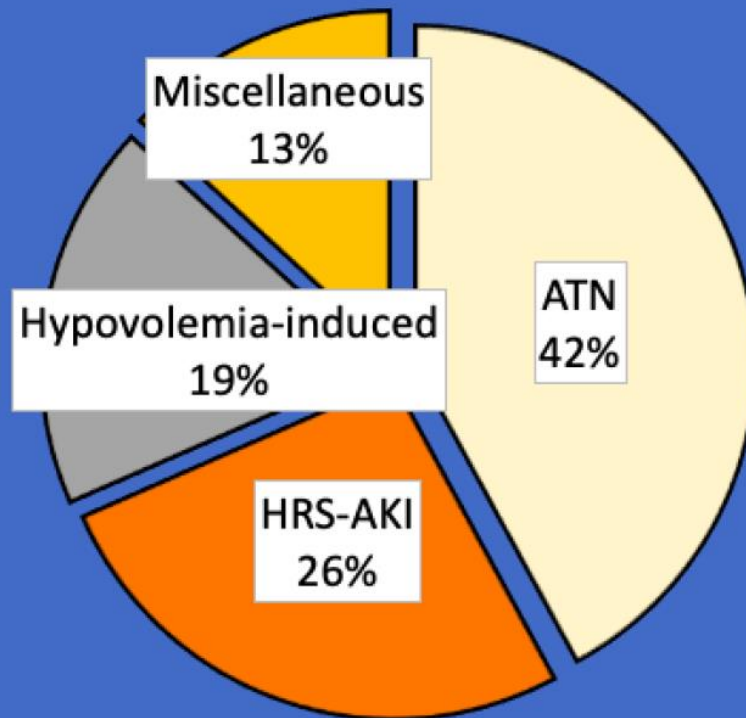
38.9% alcohol related cirrhosis
15.4% listed for transplant
8.1% received transplant



34.2% AKI stage 1
23.0% AKI stage 2
24.5% AKI stage 3
18.3% AKI stage 3 + dialysis



Phenotype of persistent AKI $\geq 1B$ after a 2-day albumin challenge



Journal Pre-proof

Prospective validation of the EASL management algorithm for acute kidney injury in cirrhosis

Ann Thu Ma, Cristina Solé, Adrià Juanola, Laia Escudé, Laura Napoleone, Emma Avitabile, Martina Pérez-Guasch, Marta Carol, Enrico Pompili, Jordi Gratacós-Ginés, Anna Soria, Ana Belén Rubio, Marta Cervera, Maria José Moreta, Manuel Morales, Elsa Solà, Esteban Poch, Núria Fabrellas, Isabel Graupera, Elisa Pose, Pere Ginès

DOI: <https://doi.org/10.1016/j.jhep.2024.03.006>

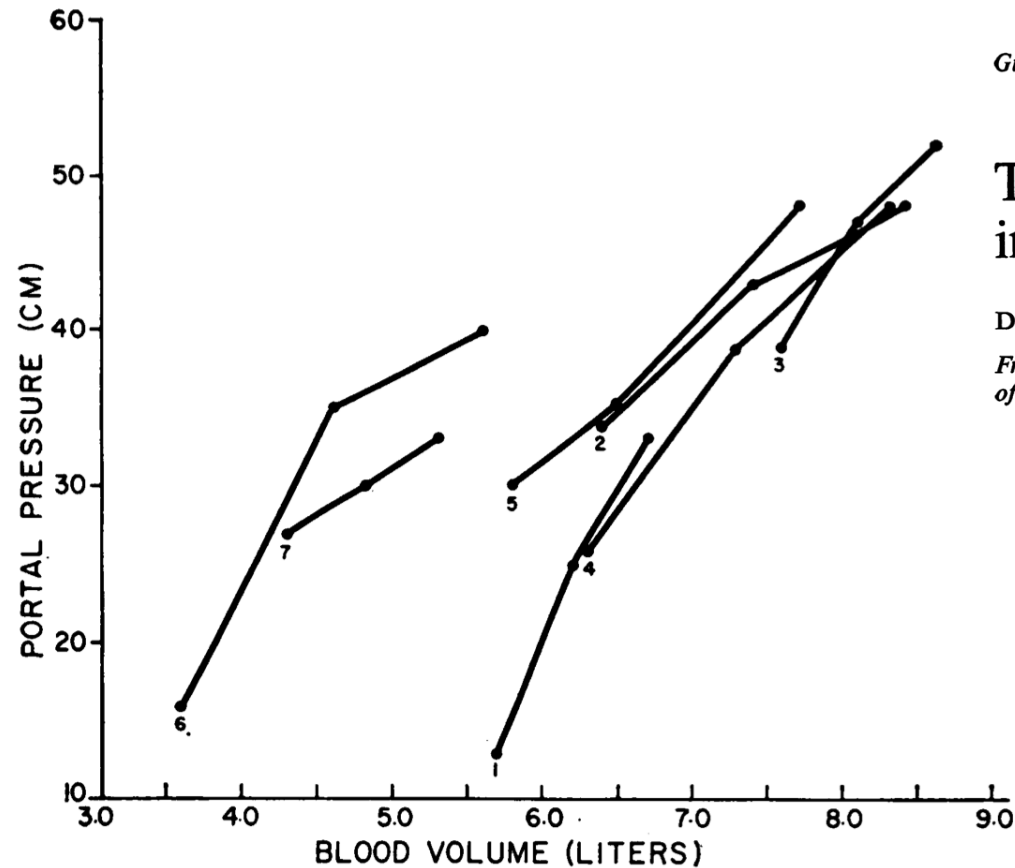
Reference: JHEPAT 9535

Accepted Date: 8 March 2024

To appear in: *Journal of Hepatology*



ALBUMINA EM IRA PÓS HDA ?



Gut, 1974, 15, 99-101

The portal pressure-blood volume relationship in cirrhosis¹

DAVID S. ZIMMON AND RICHARD E. KESSLER

From the Medical and Surgical Services, Veterans Administration Hospital, and New York University School of Medicine, New York, New York

Fig. 1 Portal pressure-blood volume curves from seven cirrhotic patients with portal hypertension derived from three measurements of portal pressure at blood volumes differing by more than 500 ml.



A randomised-controlled trial (TARGET-C) of high vs. low target mean arterial pressure in patients with cirrhosis and septic shock

Rakhi Maiwall¹, Samba Siva Rao Pasupuleti^{2,7}, Ashini Kumar Hidam³, Anupam Kumar³, Harsh Vardhan Tevethia¹, Rajan Vijayaraghavan¹, Arpita Majumdar¹, Adarsh Prasher³, Sherin Thomas⁴, Rajendra Prasad Mathur⁵, Guresh Kumar⁶, Shiv Kumar Sarin^{1,*}

Journal of Hepatology 2023. vol. 79 | 349–361



Table 2. Differences between the two randomisation groups for secondary outcome measures for intention-to-treat and per-protocol analysis.

Secondary outcome measures	Intention-to-treat analysis			Per-protocol analysis		
	Low MAP (n = 75)	High MAP (n = 75)	p value	Low MAP (n = 67)	High MAP (n = 57)	p value
AKI recovery	23 (31)	34 (45%)	0.064	21 (31)	30 (53)	0.016
Renal replacement therapy	40 (53)	43 (57%)	0.62	34 (51)	28 (49)	0.86
Intradialytic hypotension*	19/40 (48)	5/43 (12)	<0.001	18/34 (53)	1/28 (4)	<0.001

Manejo IRA x Cirrose

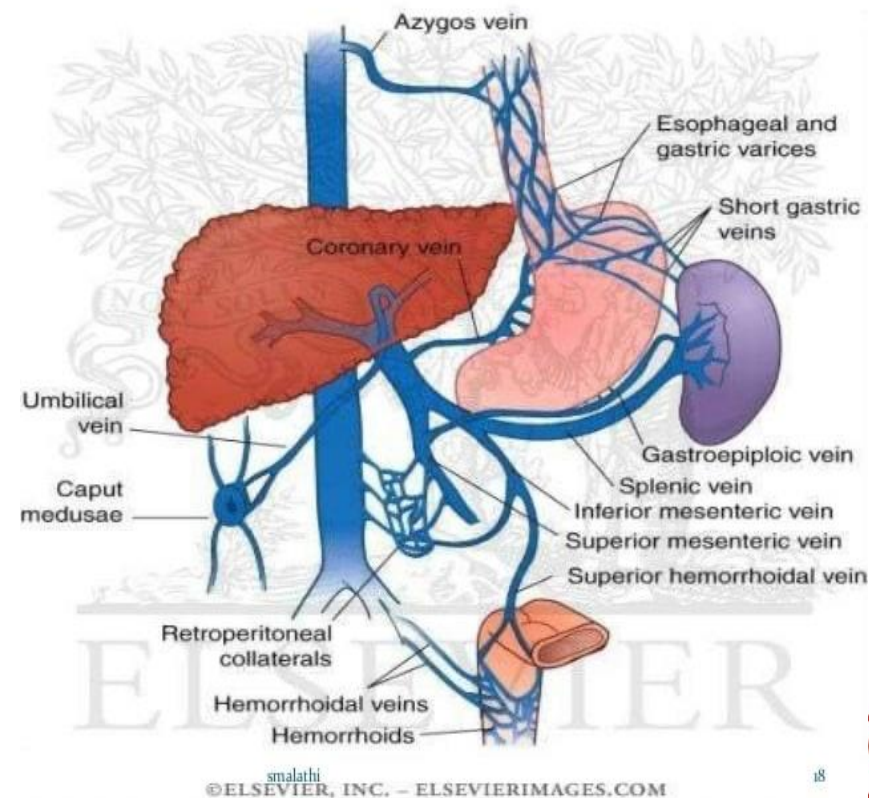
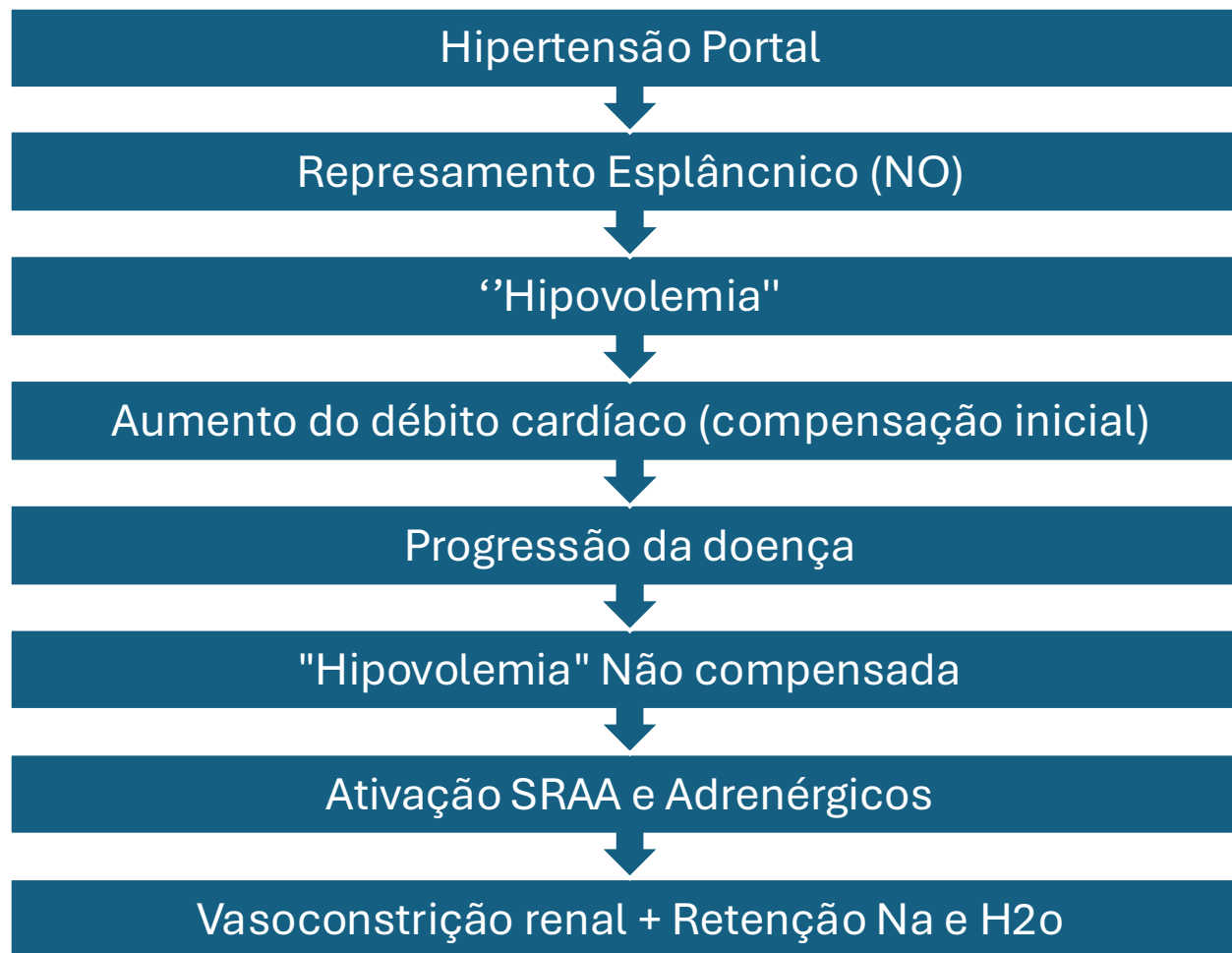
- Causa Base
- PAM ??

Cirrose + Choque Séptico + IRA

- 60-65 mmHg
- 80-85 mmHg



SÍNDROME HEPATORENAL



<https://br.pinterest.com/sergiinikol/portal-hypertension/>

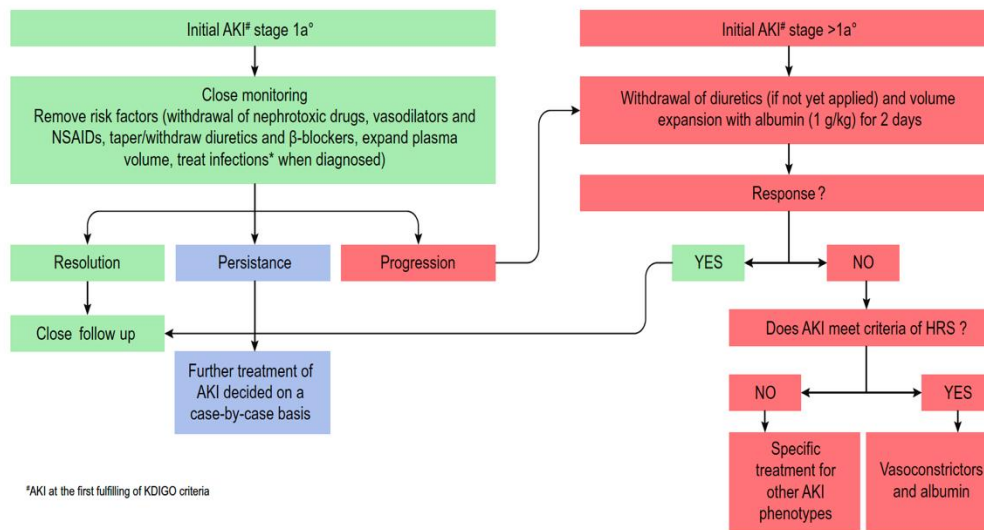


PROBLEMA COM CRITÉRIO ANTIGO

EASL Clinical Practice Guidelines for the management of patients with decompensated cirrhosis[☆]

European Association for the Study of the Liver^{*}

JOURNAL
OF HEPATOLOGY



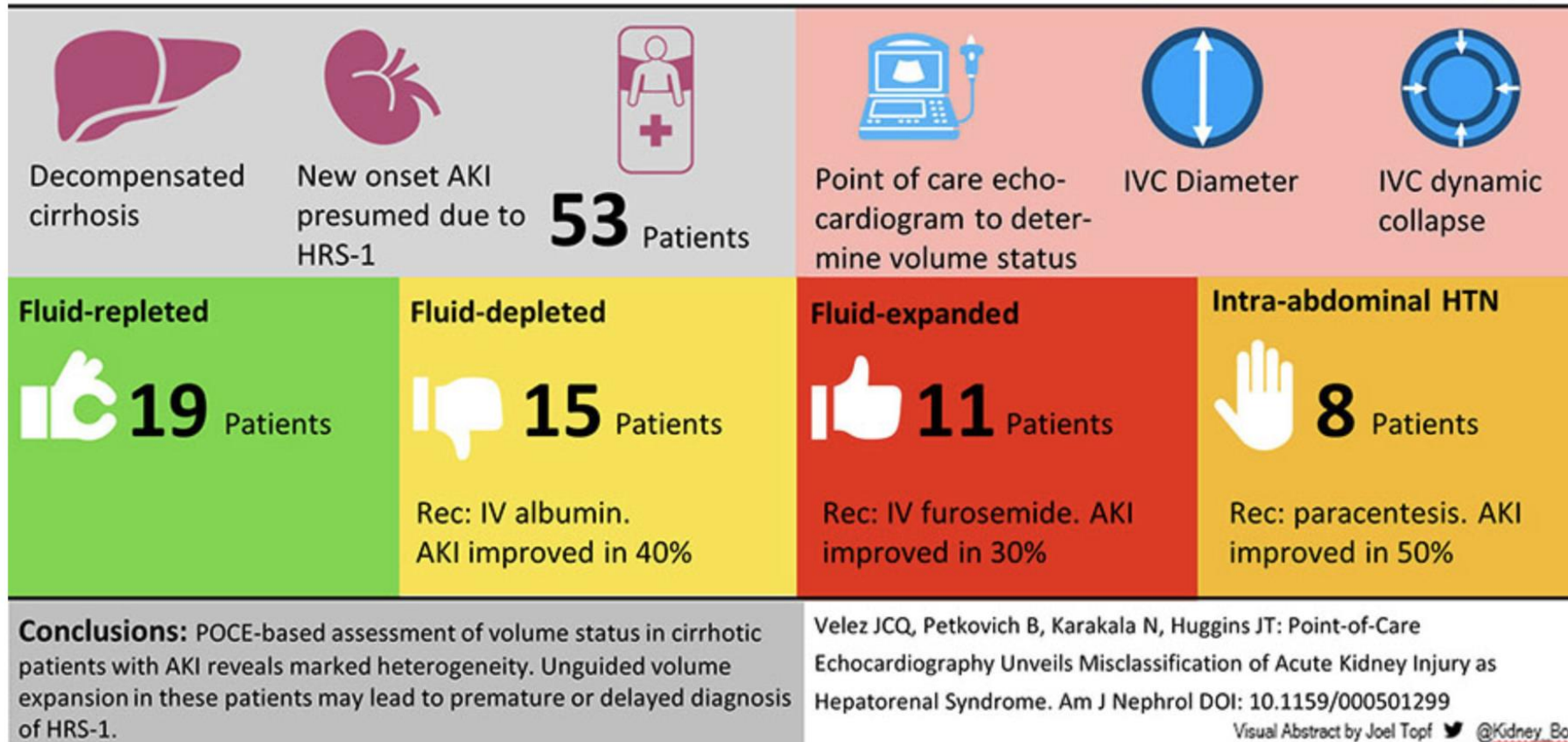
- Expandir 48h >> Avaliar SHR ?
- Avaliar SHR apenas com creatinina > 1.5 ??
- USG e urina 1 ?
- **Baixa Sensibilidade**
- SHR tipo 1 e tipo 2 ?



NEM TUDO É SHR !

Does Point of Care Ultrasound Change Management of Hepatorenal Syndrome?

Nephrology American Journal of X



PARACENTESE

Manipulation of Ascitic Fluid Pressure in Cirrhotics to Optimize Hemodynamic and Renal Function

TABLE 4. Renal Function Parameters Pre- and Postparacentesis

	Preparacentesis (51 studies) Mean $\pm$ SD		Postparacentesis (51 studies) Mean $\pm$ SD	
IAP (cm H ₂ O)	33.47 $\pm$	9.12	19.06 $\pm$	4.37*
Hematocrit (%)	32.0 $\pm$	2.22	32.33 $\pm$	1.94†
BUN (mg/dl)	27.04 $\pm$	8.99	25.3 $\pm$	8.17*
Serum Creatinine (mg/dl)	1.37 $\pm$	0.49	1.32 $\pm$	0.58*
Serum Osmolality (mOsm)	289.0 $\pm$	9.77	287.47 $\pm$	8.14‡
Serum Sodium (mEq/L)	135.27 $\pm$	4.54	135.14 $\pm$	3.88‡
Urine Osmolality (mOsm)	447.51 $\pm$	130.91	405.63 $\pm$	152.7‡
Urine Volume (cc/hour)	46.74 $\pm$	26.46	54.95 $\pm$	24.52†
Urine Sodium (mEq/L)	51.16 $\pm$	33.41	34.27 $\pm$	29.2‡
Urine Creatinine (mg/dl)	85.47 $\pm$	54.15	97.69 $\pm$	56.17*
Ccr (cc/min)	46.22 $\pm$	18.2	63.73 $\pm$	20.9*
Cosm (cc/min)	71.40 $\pm$	35.39	94.9 $\pm$	46.2*
CH ₂ O (cc/min)	0.71 $\pm$	0.26	0.67 $\pm$	0.36‡
FeNa (%)	0.72 $\pm$	0.48	0.61 $\pm$	0.37‡
RFI (%)	0.98 $\pm$	0.66	0.82 $\pm$	51.0‡

* P < 0.001. †P < 0.01. ‡Not significant.



ASCITE > PIA > DISFUNÇÃO RENAL ?

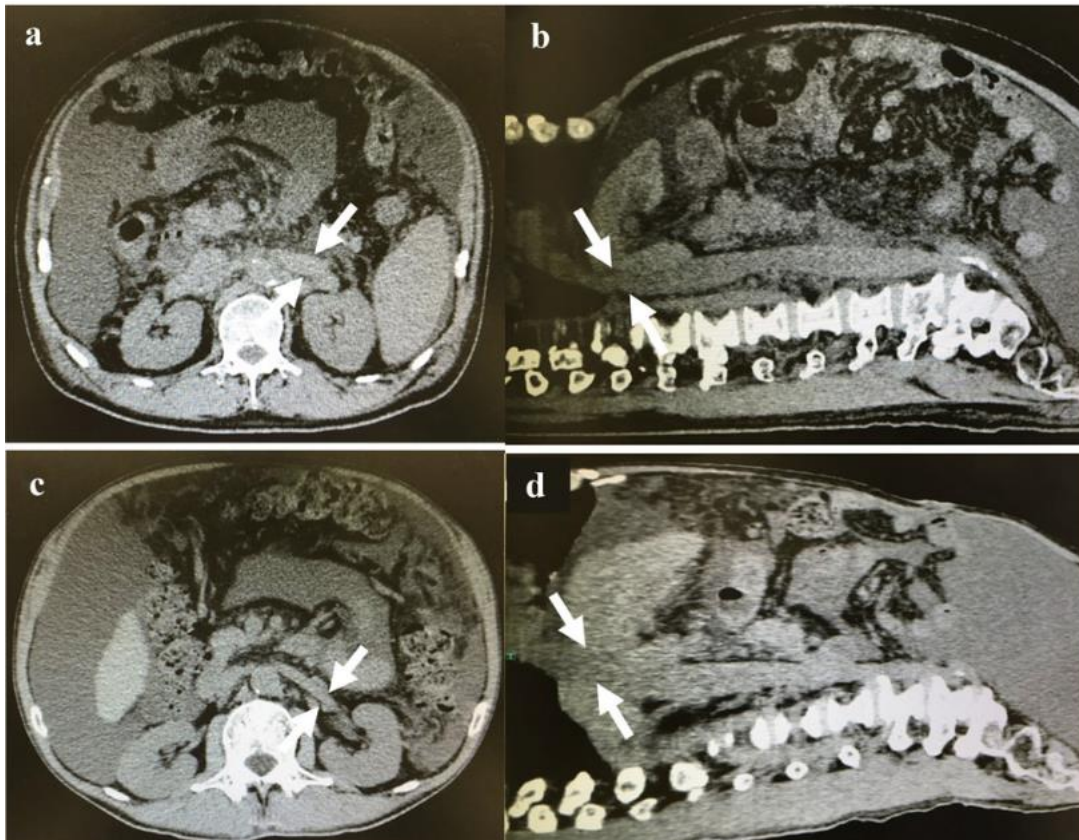
Hepatology Research 2018; 48: E117–E125

doi: 10.1111/hepr.12935

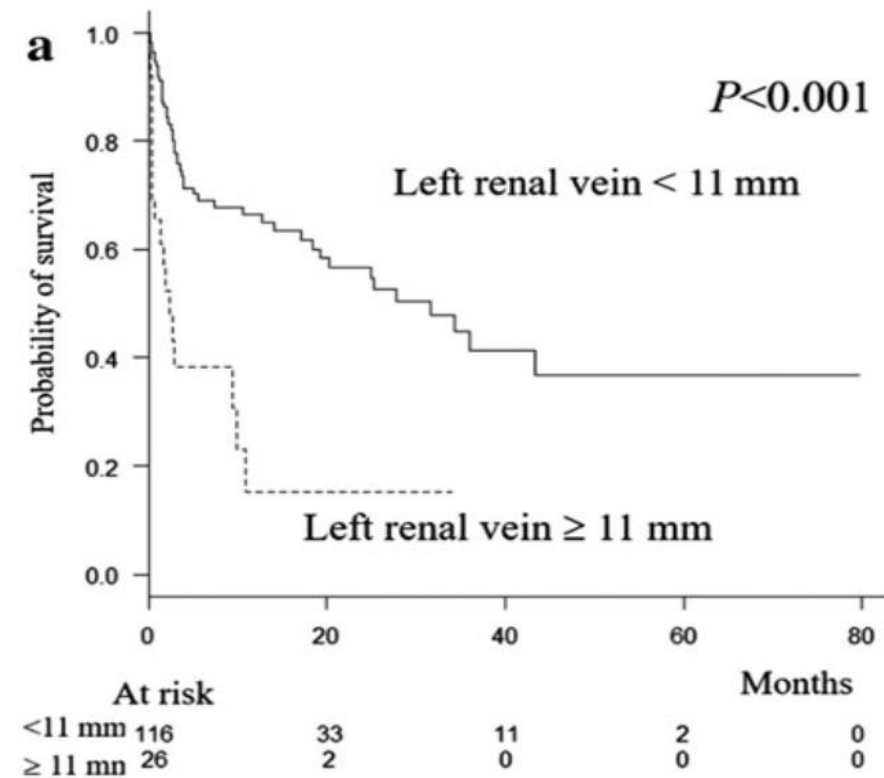
Original Article

Renal vein dilation predicts poor outcome in patients with refractory cirrhotic ascites

Naoki Matsumoto,¹ Masahiro Ogawa,¹ Mariko Kumagawa,¹ Yukinobu Watanabe,¹ Midori Hirayama,¹ Takao Miura,¹ Hiroshi Nakagawara,^{1,2} Shunichi Matsuoka,^{1,2,3} Mitsuhiro Moriyama¹ and Hirotoshi Fujikawa³



PIA ? MCP CIRROSE ? HPP?



Acute kidney injury in patients with cirrhosis: Acute Disease Quality Initiative (ADQI) and International Club of Ascites (ICA) joint multidisciplinary consensus meeting

Mitra K. Nadim¹, John A. Kellum², Lui Forni³, Claire Francoz⁴, Surmeet K. Asrani⁵, Marlies Ostermann⁶, Andrew S. Allegretti⁷, Javier A. Neyra⁸, Jody C. Olson⁹, Salvatore Piano¹⁰, Lisa B. VanWagner¹¹, Elizabeth C. Verna¹², Ayse Akcan-Arkan¹³, Paolo Angeli¹⁴, Justin M. Belcher^{15,16}, Scott W. Biggins¹⁷, Akash Deep¹⁸, Guadalupe Garcia-Tsao^{19,16}, Yuri S. Genyk^{20,21}, Pere Gines²², Patrick S. Kamath²³, Sandra L. Kane-Gill²⁴, Manish Kaushik²⁵, Nuttha Lumertgul²⁶, Etienne Macedo²⁷, Rakhi Maiwall²⁸, Sebastian Marciano²⁹, Raimund H. Pichler³⁰, Claudio Ronco³¹, Puneeta Tandon³², Juan-Carlos Q. Velez^{33,34}, Ravindra L. Mehta³⁵, François Durand^{4,36*}

Box 1. ICA-ADQI new diagnostic criteria for HRS-AKI.

- Cirrhosis with ascites
- Increase in serum creatinine ≥ 0.3 mg/dl ($26.5 \mu\text{mol/L}$) within 48 h or $\geq 50\%$ from baseline value known or presumed to have occurred within the prior 7 days and/or urinary output ≤ 0.5 ml/kg for ≥ 6 h
- Absence of improvement in serum creatinine and/or urine output within 24 h following adequate volume resuscitation (when clinically indicated)
- Absence of strong evidence for an alternative explanation as the primary cause of AKI

Presence of underlying kidney disease does not exclude a diagnosis of superimposed HRS-AKI and HRS-AKI may coexist with other causes of AKI. Examples of alternative causes of AKI include septic shock requiring vasopressors, drug-induced AKI, obstruction, or acute glomerular injury. Patients who meet HRS criteria are considered to have HRS-AKI, HRS-AKD or HRS-CKD based on timing and duration of kidney dysfunction. ADQI, Acute Disease Quality Initiative; AKD, acute kidney disease; AKI, acute kidney injury; CKD, chronic kidney disease; HRS, hepatorenal syndrome; ICA, International Club of Ascites.

NOVO CRITÉRIO (2024)

CIRROSE + ASCITE

KDIGO 1

Não melhora após 24h de expansão (sn)

Ausência de outra causa mais provável !



cohort, the median serum creatinine at the start of terlipressin therapy was of only 2.4 mg/dL, which is lower than the pretreatment serum creatinine values in most of the RCTs, including a mean of 3.5 mg/dL in patients included in the CONFIRM

Journal Pre-proof

Prospective validation of the EASL management algorithm for acute kidney injury in cirrhosis

Ann Thu Ma, Cristina Solé, Adrià Juanola, Laia Escudé, Laura Napoleone, Emma Avitabile, Martina Pérez-Guasch, Marta Carol, Enrico Pompili, Jordi Gratacós-Ginés, Anna Soria, Ana Belén Rubio, Marta Cervera, Maria José Moreta, Manuel Morales, Elsa Solà, Esteban Poch, Núria Fabrellas, Isabel Graupera, Elisa Pose, Pere Ginès

	AKI meeting HRS-AKI criteria	AKI not meeting HRS-AKI criteria
Kidney outcomes ^a	n=23 ^b	n=84
Overall response	61%	70%
Full response	48%	50%
Partial response	13%	20%

The NEW ENGLAND JOURNAL of MEDICINE

ORIGINAL ARTICLE

Terlipressin plus Albumin for the Treatment of Type 1 Hepatorenal Syndrome

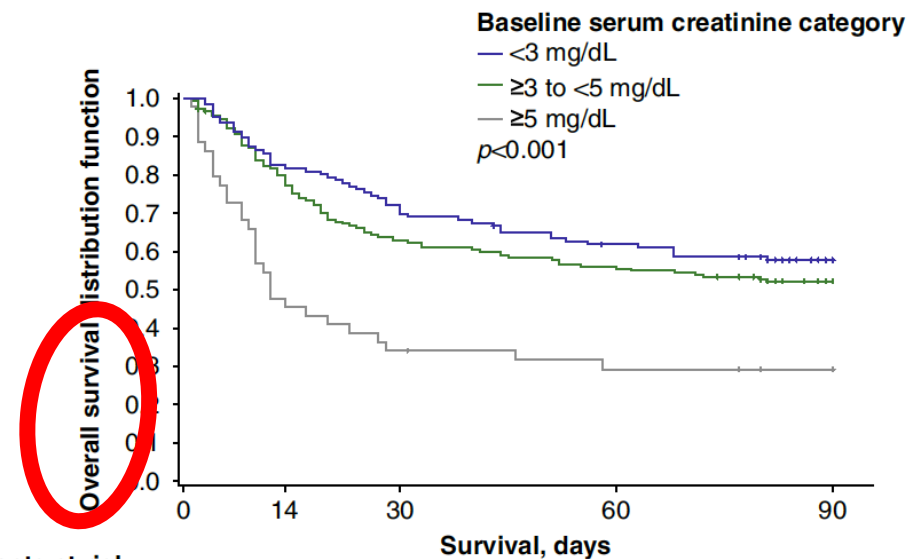
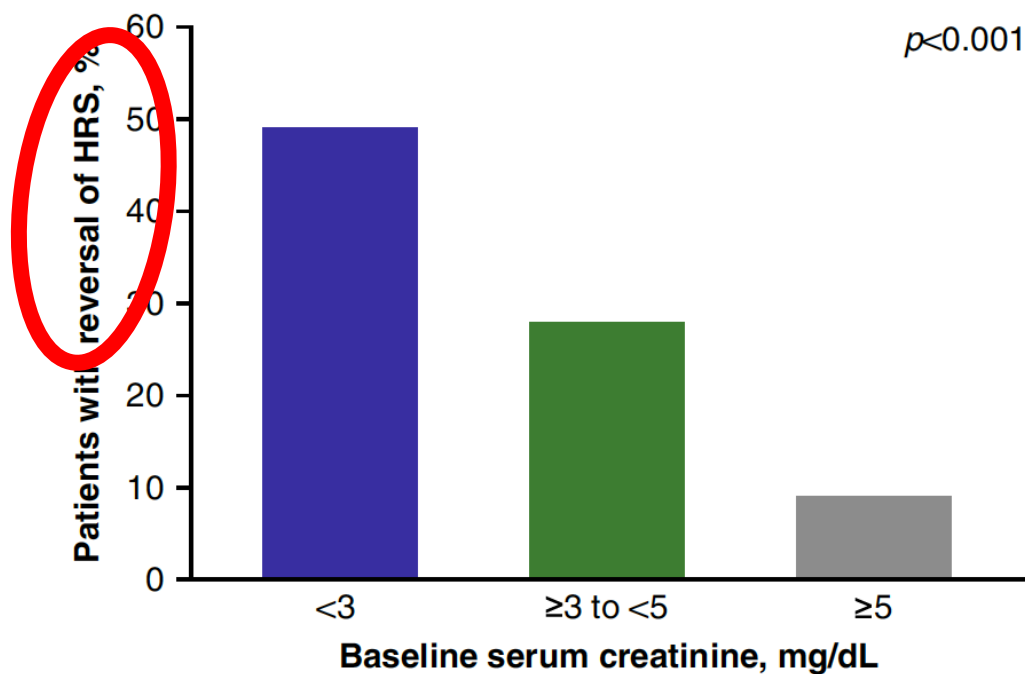
F. Wong, S.C. Pappas, M.P. Curry, K.R. Reddy, R.A. Rubin, M.K. Porayko, S.A. Gonzalez, K. Mumtaz, N. Lim, D.A. Simonetto, P. Sharma, A.J. Sanyal, M.J. Mayo, R.T. Frederick, S. Escalante, and K. Jamil, for the CONFIRM Study Investigators*

Table 2. Primary and Four Secondary End Points Included in Multiplicity Adjustment.*			
End Point	Terlipressin	Placebo	P Value
	number/total number of patients (percent)		
Primary end point of verified reversal of HRS†	63/199 (32)	17/101 (17)	0.006
Clinical success	63/199 (32)	17/101 (17)	



Early treatment with terlipressin in patients with hepatorenal syndrome yields improved clinical outcomes in North American studies

Michael P. Curry¹ | Hugo E. Vargas² | Alex S. Befeler³ | Nikolaos T. Pyrsopoulos⁴ | Vilas R. Patwardhan¹ | Khurram Jamil⁵



Patients at risk

Baseline serum creatinine	0	14	30	60	90
<3 mg/dL	126	104	91	76	52
≥3 to <5 mg/dL	182	144	113	101	77
≥5 mg/dL	44	21	15	12	10



CIRROSE + ASCITE

KDIGO 1

Não melhora após 24h de expansão (sn)

Ausência de outra causa mais provável !



TRATAR COMO SHR !



PROBLEMA DO NOVO CRITÉRIO

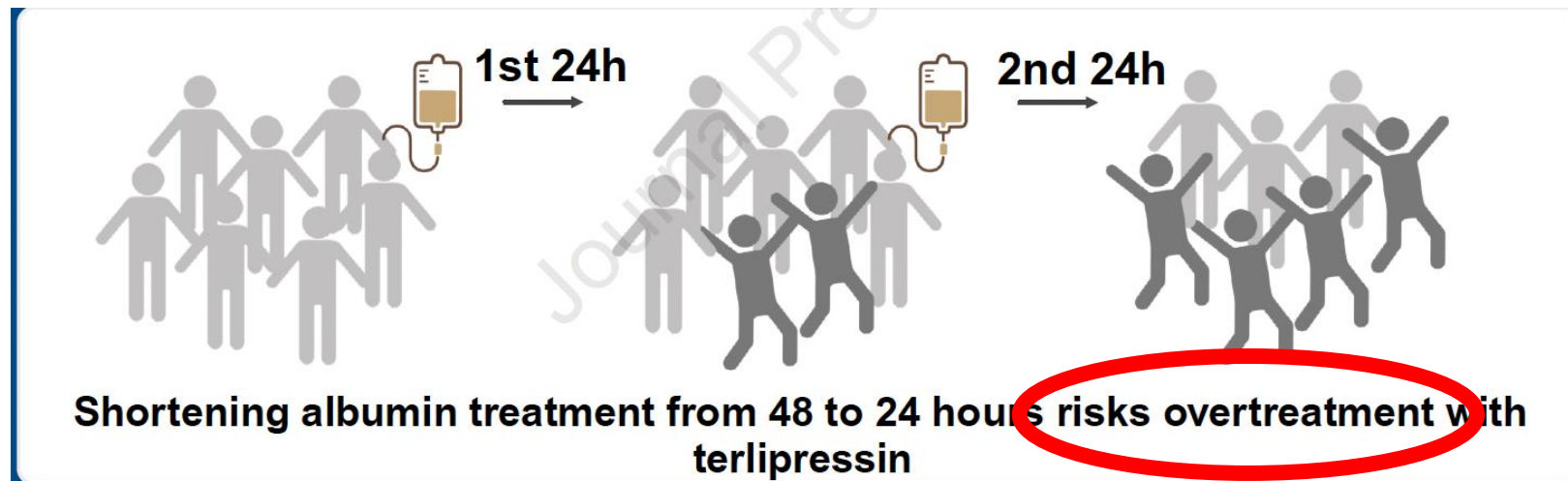
RESEARCH ARTICLE · Articles in Press, March 19, 2025 · Open Access

Effect of albumin treatment duration on response rates and outcomes in patients with cirrhosis and acute kidney injury

[Eva Maria Schleicher](#)^{1,2}  · [Henrik Karbanek](#)³  · [Julia Weinmann-Menke](#)^{1,4}  · ... · [Alexander Zipprich](#)³  · [Cristina Ripoll](#)^{3,*}  · [Christian Labenz](#)^{1,2,*}  ... [Show more](#)

[Affiliations & Notes](#) ▾ [Article Info](#) ▾ [Linked Articles \(1\)](#) ▾

We found that 38 %, 24 %, and 25 % of the cohort responded to albumin therapy during the second 24 hours of the treatment according to definitions 1, 2, and 3,



COMO TRATAR SHR ?



Albumina
20-40g/dia
Fracionar !
Cuidado com HP



Terlipressina
Bolus x BIC
0.5-2mg 6/6h



O QUE A ALBUMINA FAZ ?

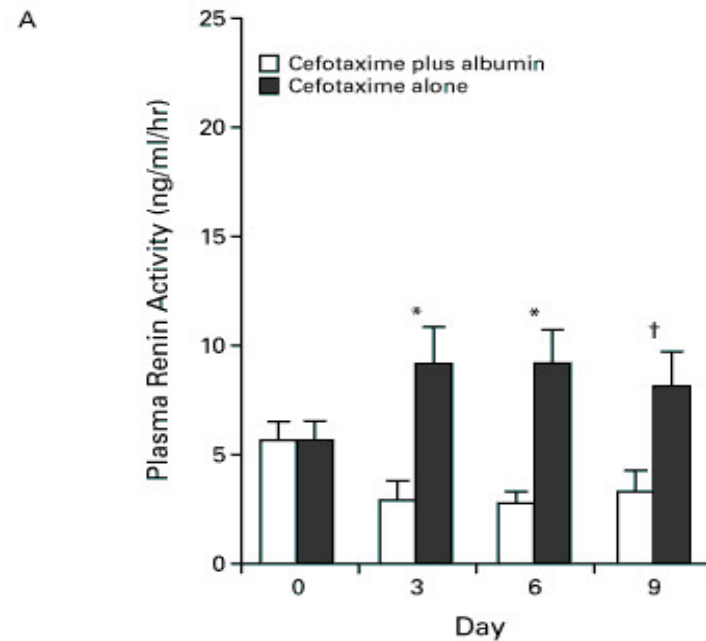
ORIGINAL ARTICLE

f x in e

Effect of Intravenous Albumin on Renal Impairment and Mortality in Patients with Cirrhosis and Spontaneous Bacterial Peritonitis

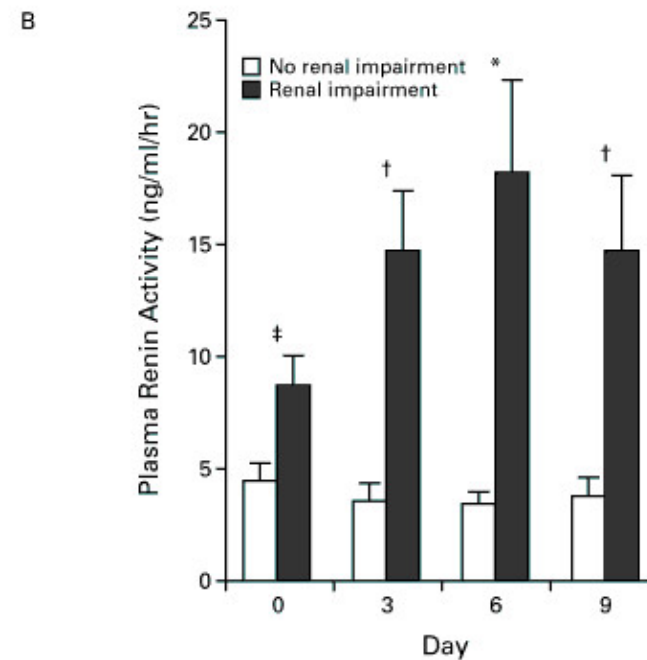
Authors: Pau Sort, M.D., Miquel Navasa, M.D., Vicente Arroyo, M.D., Xavier Aldeguer, M.D., Ramon Planas, M.D., Luis Ruiz-del-Arbol, M.D., Lluís Castells, M.D., Víctor Vargas, M.D., Germán Soriano, M.D., Mónica Guevara, M.D., Pere Ginès, M.D., and Joan Rodés, M.D. [Author Info & Affiliations](#)

Published August 5, 1999 | N Engl J Med 1999;341:403-409 | DOI: 10.1056/NEJM199908053410603
VOL. 341 NO. 6



No. of PATIENTS

Cefotaxime plus albumin	63	60	58	57
Cefotaxime alone	63	60	56	51



No. of PATIENTS

No renal impairment	99	96	96	94
Renal impairment	27	24	18	14



ORIGINAL ARTICLE

Terlipressin plus Albumin for the Treatment of Type 1 Hepatorenal Syndrome

F. Wong, S.C. Pappas, M.P. Curry, K.R. Reddy, R.A. Rubin, M.K. Porayko, S.A. Gonzalez, K. Mumtaz, N. Lim, D.A. Simonetto, P. Sharma, A.J. Sanyal, M.J. Mayo, R.T. Frederick, S. Escalante, and K. Jamil, for the CONFIRM Study Investigators*

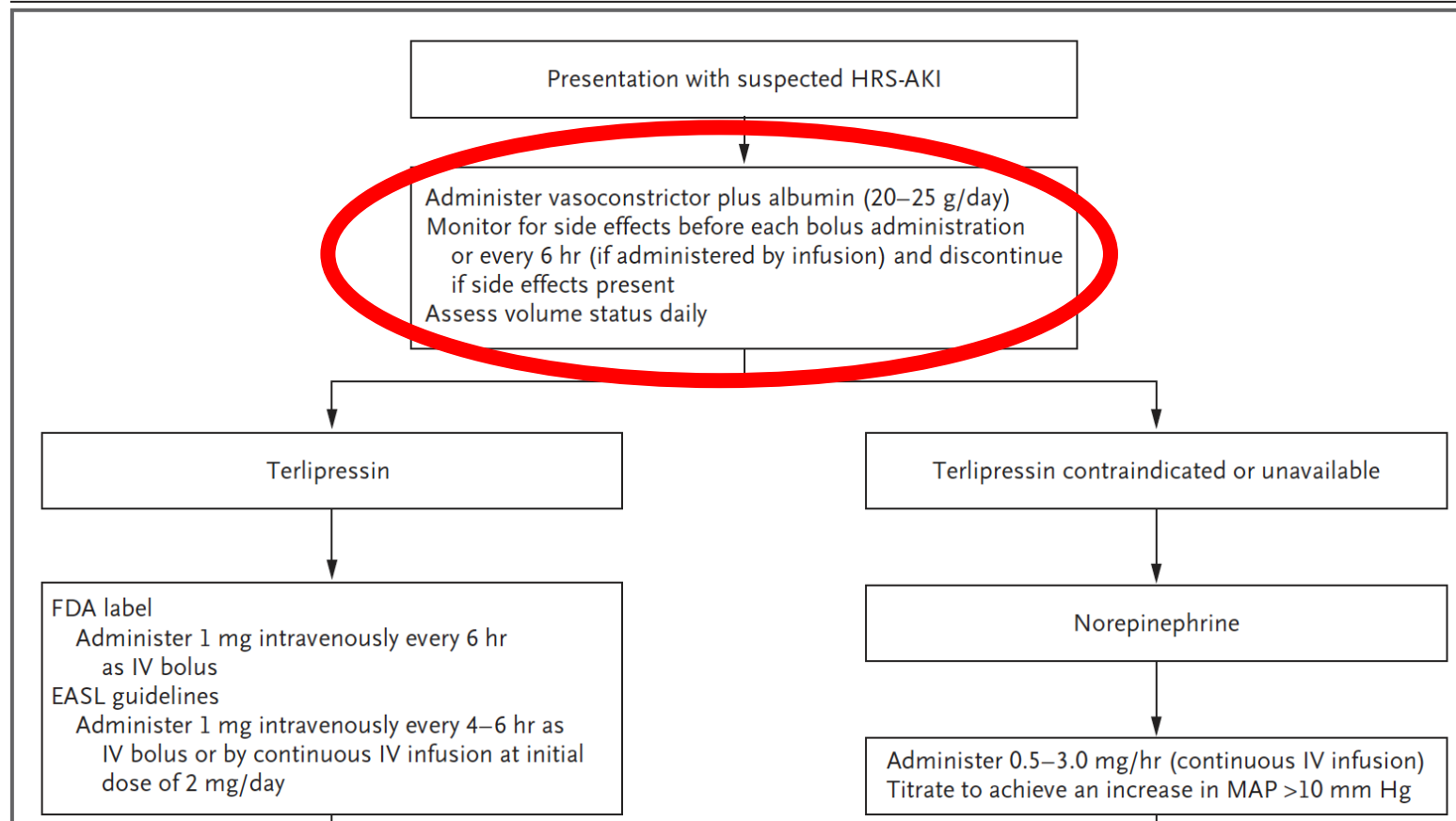
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Primary end point of verified reversal of HRS†			0.006
Clinical success	63/199 (32)	17/101 (17)	

Table 4. Adverse Events in the Safety Population.*		
Event	Terlipressin (N = 200)	Placebo (N = 99)
	<i>number of patients (percent)</i>	
Respiratory failure	20 (10)	3 (3)



Acute Kidney Injury in Patients with Cirrhosis

Mitra K. Nadim, M.D., and Guadalupe Garcia-Tsao, M.D.



Terlipressin Given by Continuous Intravenous Infusion Versus Intravenous Boluses in the Treatment of Hepatorenal Syndrome: A Randomized Controlled Study

Marta Cavallin,¹ Salvatore Piano,^{1,2} Antonietta Romano,¹ Silvano Fasolato,^{1,2} Anna Chiara Frigo,³ Gianpiero Benetti,⁴ Elisabetta Gola,¹ Filippo Morando,¹ Marialuisa Stanco,¹ Silvia Rosi,¹ Antonietta Sticca,¹ Umberto Cillo,⁵ and Paolo Angeli^{1,2}

TABLE 3. Severe Treatment-Related Adverse Events (Defined as Needed to Withdraw Terlipressin)

	TERLI-INF Group (n = 34)	TERLI-BOL Group (n = 37)	P
Patients, no. (%)	7 (20.59)	16 (43.24)	<0.05
Suspected intestinal ischemia	—	3	—
Peripheral ischemia	1	—	—
Circulatory overload	2	5	—
Angina pectoris	3	3	—
Arrhythmia	—	1	—
Arterial hypertension	1	—	—
Persistent diarrhea	—	2	—
Others	—	2	—

Abbreviations: TERLI-BOL, terlipressin by intravenous boluses; TERLI-INF, terlipressin by intravenous infusion.

latter approach. In this regard, it should be noted that 78.95% of complete responders in the TERLI-INF group did it at a dose of 2 mg/day, a dose lower than the starting dose in the TERLI-BOL group. These



80 % Respondeu
com dose menor !



TERLIPRESSINA X NORADRENALINA

Terlipressin versus noradrenaline in the treatment of hepatorenal syndrome: systematic review with meta-analysis and full economic evaluation

Ângelo Z. de Mattos^{a,b,c,d}, Angelo A. de Mattos^a and Rodrigo A. Ribeiro^{d,e,f}

350 European Journal of Gastroenterology & Hepatology

March 2016 • Volume 28 • Number 3

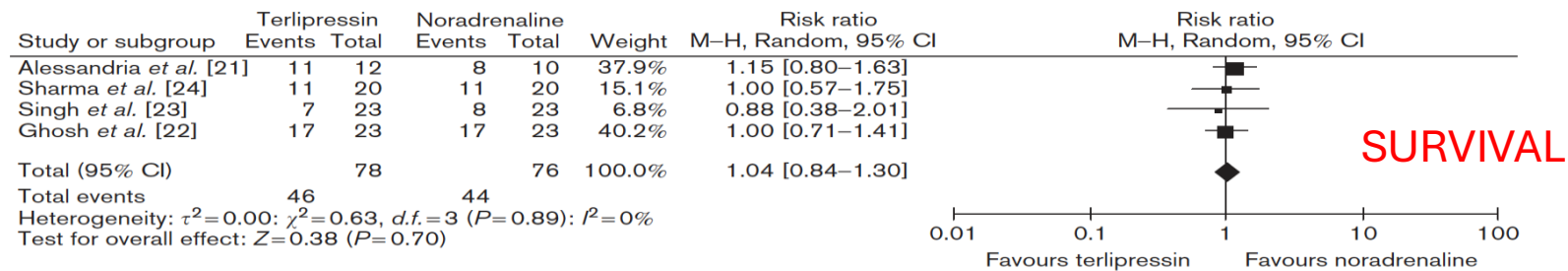


Fig. 2. Forest plot of the random-effects model meta-analysis for the comparison between terlipressin and noradrenaline in terms of 30-day survival. Each study is identified by the name of the first author and reference number. Squares indicate the risk ratio, and their sizes are proportional to the weights of the trials. The horizontal bars refer to the 95% confidence interval (CI) of the risk ratios. The vertical line is the equivalence line, where the risk ratio is 1. The diamond represents the 95% CI of the pooled risk ratio. M-H, Mantel-Haenszel method.

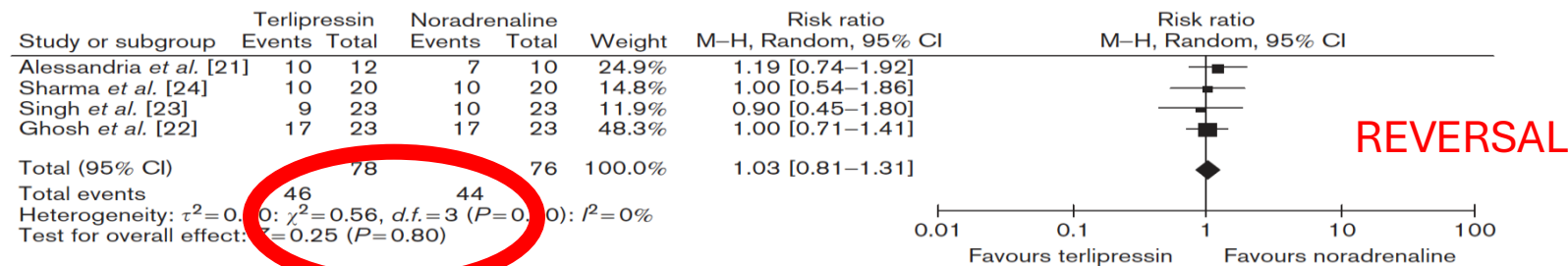


Fig. 3. Forest plot of the random-effects model meta-analysis for the comparison between terlipressin and noradrenaline in hepatorenal syndrome reversal. Each study is identified by the name of the first author and reference number. Squares indicate the risk ratio and their sizes are proportional to the weights of the trials. The horizontal bars refer to the 95% confidence interval (CI) of the risk ratios. The vertical line is the equivalence line, where the risk ratio is 1. The diamond represents the 95% CI of the pooled risk ratio. M-H, Mantel-Haenszel method.



TERLIPRESSINA X NORADRENALINA

HEPATOLOGY, VOL. 0, NO. 0, 2019

Terlipressin Is Superior to Noradrenaline in the Management of Acute Kidney Injury in Acute on Chronic Liver Failure

Vinod Arora,¹ Rakhi Maiwall,¹ Vijayaraghavan Rajan,¹ Ankur Jindal,¹ Saggere Muralikrishna Shasthry,¹ Guresh Kumar,² Priyanka Jain,² and Shiv Kumar Sarin¹

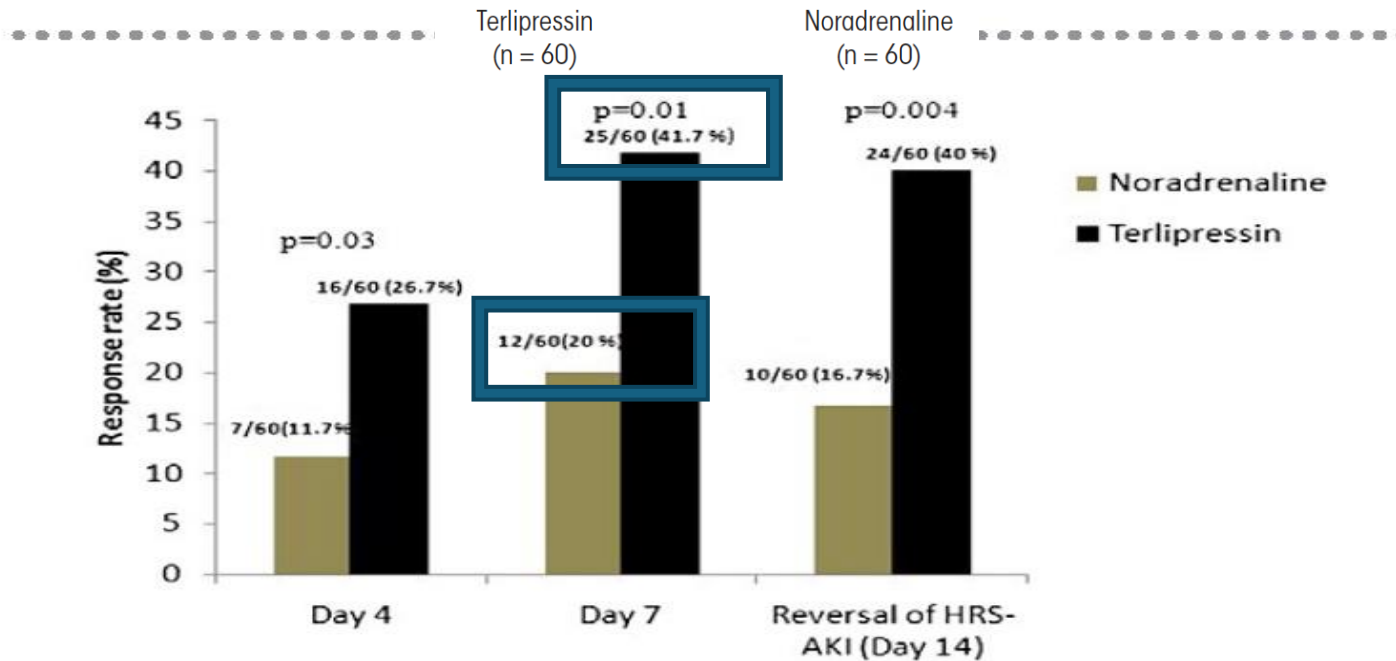


FIG. 2. Rate of response (complete and partial) in patients who were randomized to terlipressin and noradrenaline arms.



LEMBRAR DA PROFILAXIA NA PBE!

ORIGINAL ARTICLE

f X in

Effect of Intravenous Albumin on Renal Impairment and Mortality in Patients with Cirrhosis and Spontaneous Bacterial Peritonitis

Authors: Pau Sort, M.D., Miquel Navasa, M.D., Vicente Arroyo, M.D., Xavier Aldeguer, M.D., Ramon Planas, M.D., Luis Ruiz-del-Arbol, M.D., Lluís Castells, M.D., Victor Vargas, M.D., Germán Soriano, M.D., Mónica Guevara, M.D., Pere Ginès, M.D., and Joan Rodés, M.D. [Author Info & Affiliations](#)

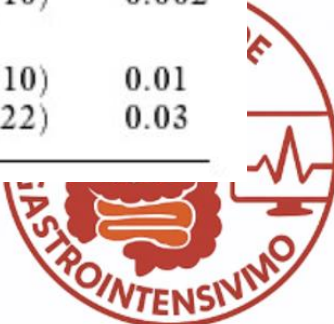
Published August 5, 1999 | N Engl J Med 1999;341:403-409 | DOI: 10.1056/NEJM199908053410603
VOL. 341 NO. 6

mg per deciliter. In patients assigned to receive cefotaxime and albumin, albumin (Albúmina 20 percent, Instituto Grífols, Barcelona, Spain) was given at a dose of 1.5 g per kilogram of body weight during the first six hours after enrollment, followed by 1 g per kilogram on day 3 (laboratory measurements before the initiation of therapy and on day 3 were performed

→ 1.5g/kg no d1
→ 1g/kg no d3

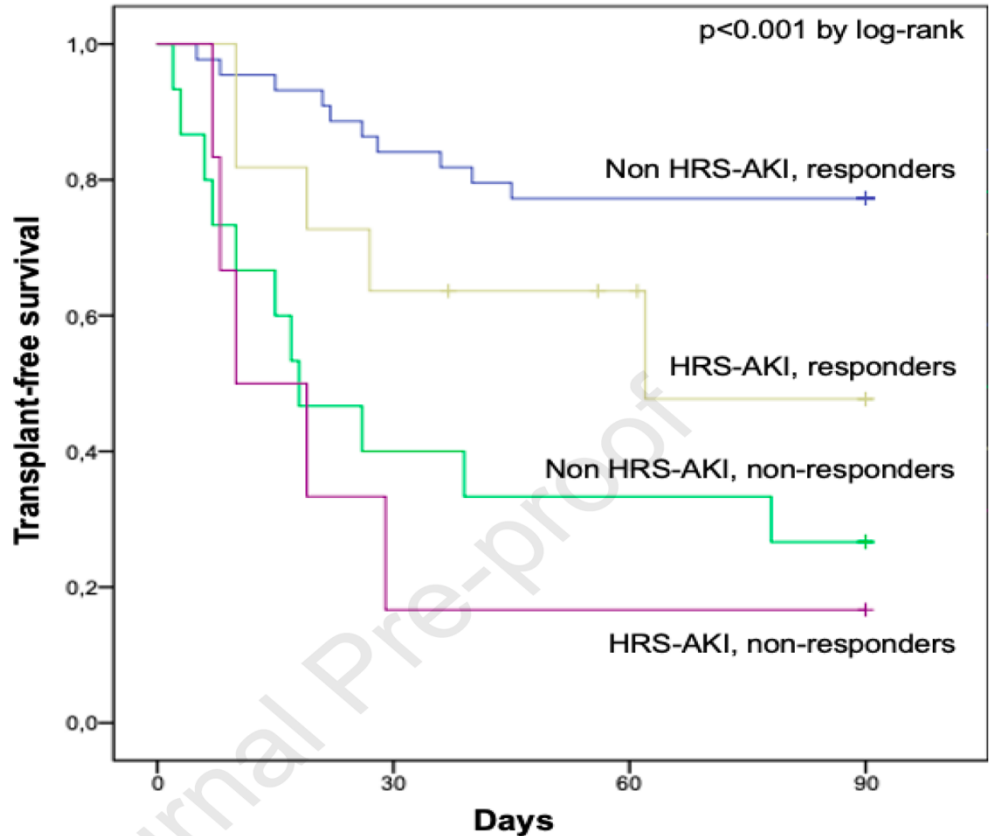
TABLE 2. CLINICAL OUTCOME ACCORDING TO THE ASSIGNED TREATMENT.*

OUTCOME VARIABLE	CEFOTAXIME (N=63)	CEFOTAXIME PLUS ALBUMIN (N=63)	P VALUE
Resolution of infection — no. (%)†	59 (94)	62 (98)	0.36
Duration of antibiotic therapy — days	6 ± 1	5 ± 1	0.48
Paracentesis for ascites after resolution of infection — no. (%)‡	16 (25)	14 (22)	0.83
Hospital stay — days	13 ± 1	14 ± 1	0.48
Renal impairment — no. (%)	21 (33)	6 (10)	0.002
Death — no. (%)			
In hospital§	18 (29)	6 (10)	0.01
At three months¶	26 (41)	14 (22)	0.03



Prospective validation of the EASL management algorithm for acute kidney injury in cirrhosis

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DOI: <https://doi.org/10.1016/j.jhep.2024.03.006>
Reference: JHEPAT 9535
Accepted Date: 8 March 2024
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Management of hepatorenal syndrome in liver cirrhosis: a recent update

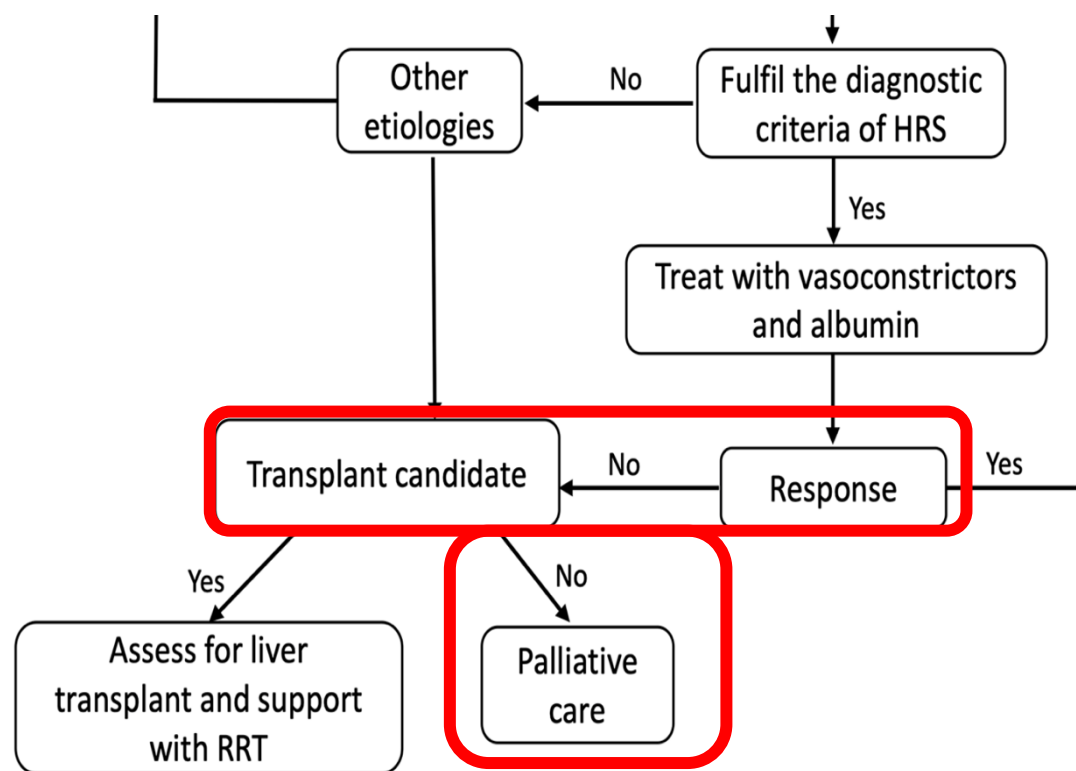
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REVERSÃO 32 – 44% (TERLI + ALB)

O QUE FAZER EM NÃO RESPONDEDORES ?

DIALISAR ?

Figure 2. Suggested management algorithm for acute kidney injury including hepatorenal syndrome.



Ascites Increases the Resting Energy Expenditure in Liver Cirrhosis

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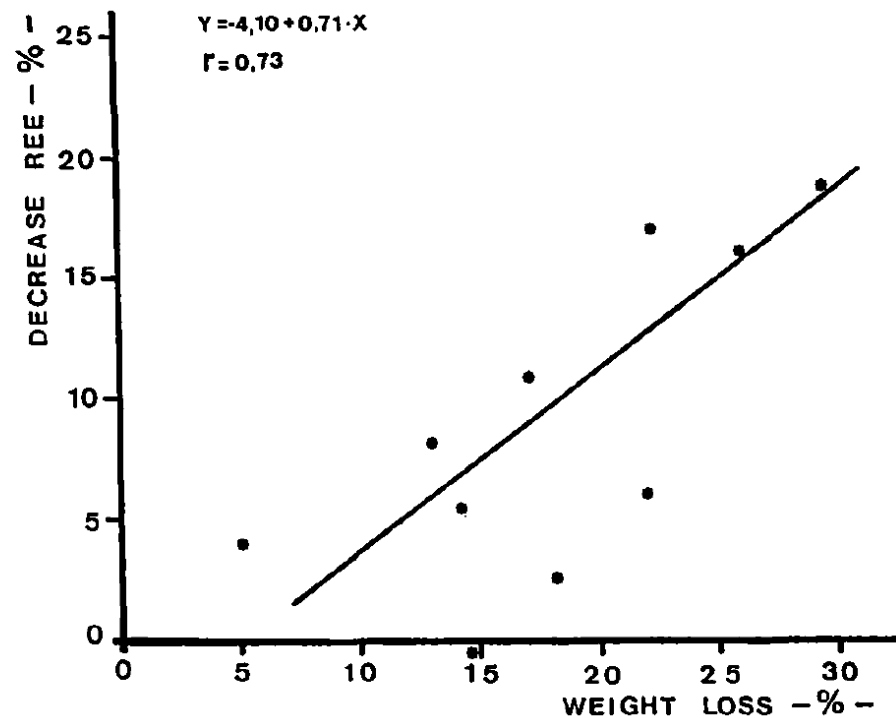


Figure 2. Relationship between the weight loss and the reduction of measured REE induced by the removal of ascites.

Paracentesis induced a mean percentage decrease of REE of 9.1 ± 6 accompanied by a mean percentage



Hepatic Encephalopathy

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reducing intestinal ammonia production.⁵⁴ Protein restriction is ill advised, since normal protein intake does not appear to exacerbate hepatic encephalopathy.^{55,56} Guidelines recommend lactu-

A **restrição de proteínas não é recomendada**, pois a ingestão normal de proteínas **não parece agravar** a encefalopatia hepática.

